

Individual Addendum: Ashley Seale **Name:** Ashley Seale **Team:** ILOVECODING **Date:** May 1, 2026 --- ## 1. Personal Contribution My contributions to the ILOVECODING capstone project were distributed evenly across all four milestones, with specific focus on data pipeline development, exploratory analysis, and final synthesis into business recommendations. - **Milestone 1 (Data Pipeline & Merge):** Designed and implemented REIT-FRED data merge architecture; validated referential integrity across 34,121 observations; implemented forward-fill strategy for quarterly leverage reporting in monthly analysis; verified data quality (0.4% loss acceptable, 98.4% retained). Created data summary tables for M1 report. **14 hours** - **Milestone 2 (Exploratory Data Analysis):** Conducted lag-correlation analysis across leverage lags 0-6 months; created quarterly return distribution charts by REIT sector; diagnosed heteroskedasticity patterns in monthly returns; drafted methodology and interpretation sections of M2 findings report. **13 hours** - **Milestone 3 (Econometric Modeling):** Implemented two-way fixed effects specification in PanelOLS with entity and time fixed effects; specified and tested entity-level clustering; created diagnostic plotting suite (residuals, Q-Q plots, heteroskedasticity tests); formatted regression tables for publication (CSV + Excel with significance stars). **14 hours** - **Milestone 4 (Investment Memo & Final Synthesis):** Co-authored Executive Summary with key findings and specific sector recommendations; drafted Methodology section with plain-English variable definitions; synthesized M1-M3 results into economic interpretations for portfolio managers; compiled final memo following template structure; created PDF and HTML submission formats. **14 hours** **Total Capstone Hours:** ~55 hours --- ## 2. One Defended Methodological Decision *Decision: Use entity-level clustering for standard errors rather than time-level clustering or no clustering.* **Reasoning:** In Milestone 3, we estimated two-way fixed effects models with 273 REITs observed across 299 months, generating 33,573 firm-time observations. The key clustering choice was whether to cluster by entity or by time. Our rationale: - **Entity-level clustering addresses serial correlation.** Within-REIT returns exhibit autocorrelation ($\rho \approx 0.15-0.25$ across lag 1), a well-documented phenomenon in financial data. Monthly shocks to a REIT's property values, capital structure, or market sentiment persist beyond a single month. Entity-level clustering corrects for this within-REIT correlation. - **Cross-sectional clustering less appropriate.** Time-level clustering would address common shocks affecting all REITs simultaneously (e.g., Fed announcements), but our model includes time fixed effects (δ_t) that already absorb aggregate monthly shocks. Double-correcting via clustering would be redundant. - **Evidence from Milestone 2:** Autocorrelation analysis revealed significant within-REIT persistence (ACF spike at lag 1) but moderate cross-sectional correlation (contemporaneous returns weakly correlated across REITs, $\rho \approx 0.12$), confirming entity-level clustering is the binding constraint. This choice is conservative for our main finding (beta coefficient): entity clustering produces larger standard errors

than no clustering, so our beta result ($t=4.46$ even with conservative clustering) is robust. For the weaker leverage result, entity clustering reveals the instability we highlight in caveats. --- ## 3. One Key Limitation

***Limitation:** Leverage is measured using accounting debt-to-assets, which excludes off-balance-sheet financing and derivatives hedging.

****Why It Matters:**** REITs use derivatives (interest rate swaps, caps, floors) extensively to hedge variable-rate debt exposure, but these do not appear on the balance sheet. A REIT with 50% accounting leverage but 100% of debt hedged faces different interest-rate sensitivity than a REIT with 50% accounting leverage and no hedges. Our leverage measure cannot distinguish between true economic exposure and hedged exposure.

****Impact on Findings:**** - This measurement error likely ****biases leverage coefficients toward zero****. If hedging is correlated with firm size or market conditions, we underestimate heterogeneity in interest-rate sensitivity. - Our ****beta findings are less affected**** because systematic risk (correlation with market returns) partially captures true economic leverage after accounting for hedging; market prices reflect total exposure including derivative positions. - Our ****leverage null finding is actually conservative:**** if anything, we should be finding smaller effects than true effects due to measurement error.

****How to Address:**** Future analysis incorporating (a) derivative positions from REIT financial disclosures, and (b) the notional value of hedged debt, would improve leverage measurement and potentially reveal more stable effects. This limitation is why our memo appropriately cautions against leverage-based tactical trades—the finding is empirically fragile and theoretically suspect due to unobserved hedging. --- ## 4. AI Audit Notes

****Task 1: Markdown memo template structure recommendations**** - ****Prompt:**** "Suggest a professional structure for a final investment committee memo including sections for executive summary, methodology, results, conclusions, and references." - ****Output:**** AI provided outline with 6+ sections, phrasing examples, and citation guidance. - ****Verification:**** Compared output against course assignment requirements (M4_RUBRIC.md) and confirmed all required sections present. - ****Changes:**** Adopted the structure but rewrote substantive content (results interpretation, recommendations) with actual M1–M3 data and analysis. AI provided only skeletons, not numerical findings.

****Task 2: Regression table formatting script**** - ****Prompt:**** "Write Python code to format regression coefficients with standard errors in parentheses, calculate t-stats from coef/SE, determine significance stars based on p-values, and export to CSV." - ****Output:**** AI produced `format\_regression\_tables.py` with mostly correct logic, but had a bug: was checking `if t\_stat > 1.96` for significance instead of comparing actual p-values. - ****Verification:**** Tested script on actual M3 regression output (beta coefficient 0.0061, SE 0.0014). Manually calculated $t\text{-stat} = 0.0061/0.0014 = 4.36$ (close to 4.46 from regression software; small differences due to rounding). Verified $p\text{-value} < 0.001$ corresponds correctly to $t > 3.29$ threshold. Found discrepancy and corrected. - ****Changes:**** Fixed p-value logic to use actual regression output p-values instead of deriving from t-stats; verified 5 key coefficients by hand. ****Task**

3: Variable definitions in plain English** - **Prompt:** "Rewrite econometric variable definitions in clear business language without jargon, suitable for portfolio managers." - **Output:** AI initially provided definitions like "CAPM beta reflects equity market factor loading" which still contained jargon. - **Verification:** Compared to portfolio manager audience expectations; tested readability with non-economist eye. - **Changes:** Rewrote to "Beta = sensitivity to market moves ($\beta=1.5$ = 50% more volatile than market)" and added practical examples like "a REIT with beta=1.5 earns 3.7% higher returns than a market-beta REIT." **Overall Assessment:** AI accelerated structural and formatting work (templates, table coding) but required substantive human review and correction. All numerical findings verified independently against regression software outputs. No AI-generated numerical claims appear in final memo without manual verification. --- **Status:** ✓ COMPLETE **Submission Date:** May 1, 2026 by 11:59 PM EST